

Introduction to Applied Cryptography – LDACS1310

Exercises Session 2

Questions

Question 1: Leaky PRG

Let $G : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$ be a pseudo-random generator.

Using G as a building block, we build the PRG G_1 as follow:

$G : \{0, 1\}^n \rightarrow \{0, 1\}^{2kn}$, for some k

$s_0 \leftarrow_{\$} \{0, 1\}^n$

for $i = 0, \dots, k - 1$:

$G(s_i) = s_{i+1} \parallel s'_{i+1}$ where $|s_{i+1}| = |s'_{i+1}|$

$= t_{i+1}$

Output $t = t_1 \parallel \dots \parallel t_k$

Show that this PRG is not secure by constructing an attack.

Question 2: Leaky PRG

Let's fix the PRG from the previous exercise, with $G : \{0, 1\}^n \rightarrow \{0, 1\}^{2n}$ be a pseudo-random generator.

Using G as a building block, we build the PRG G_2 as follow:

$G : \{0, 1\}^n \rightarrow \{0, 1\}^{kn}$, for some k

$s_0 \leftarrow_{\$} \{0, 1\}^n$

for $i = 0, \dots, k - 1$:

$G(s_i) = s_{i+1} \parallel s'_{i+1}$ where $|s_{i+1}| = |s'_{i+1}|$

$= t_{i+1}$

Output $t = s'_1 \parallel \dots \parallel s'_k$

We keep part of the output of G as a hidden internal seed s_i to generate more random. Note that by doing so, we produce an output half the size of G_1 .

Proof that G_2 is secure for $k = 2$, i.e assuming a distinguisher for G_2 exists, craft an attack on G . *Note that it can be proven for greater k but the proof is more involved.*

Question 3:

Let G be a pseudorandom generator with expansion factor $\ell(n) > 2n$. In each of the following cases, say whether G' is necessarily a pseudorandom generator. If yes, give a proof; if not, show a counterexample.

1. Define $G'(s) = G(s_1 \cdots s_{\lceil n/2 \rceil})$, where $s = s_1 \cdots s_n \in \{0, 1\}^n$.
2. Define $G'(s) = G(0^{|s|} \parallel s)$ where $s \in \{0, 1\}^n$, that is, we prepend $|s|$ '0' bits to s .